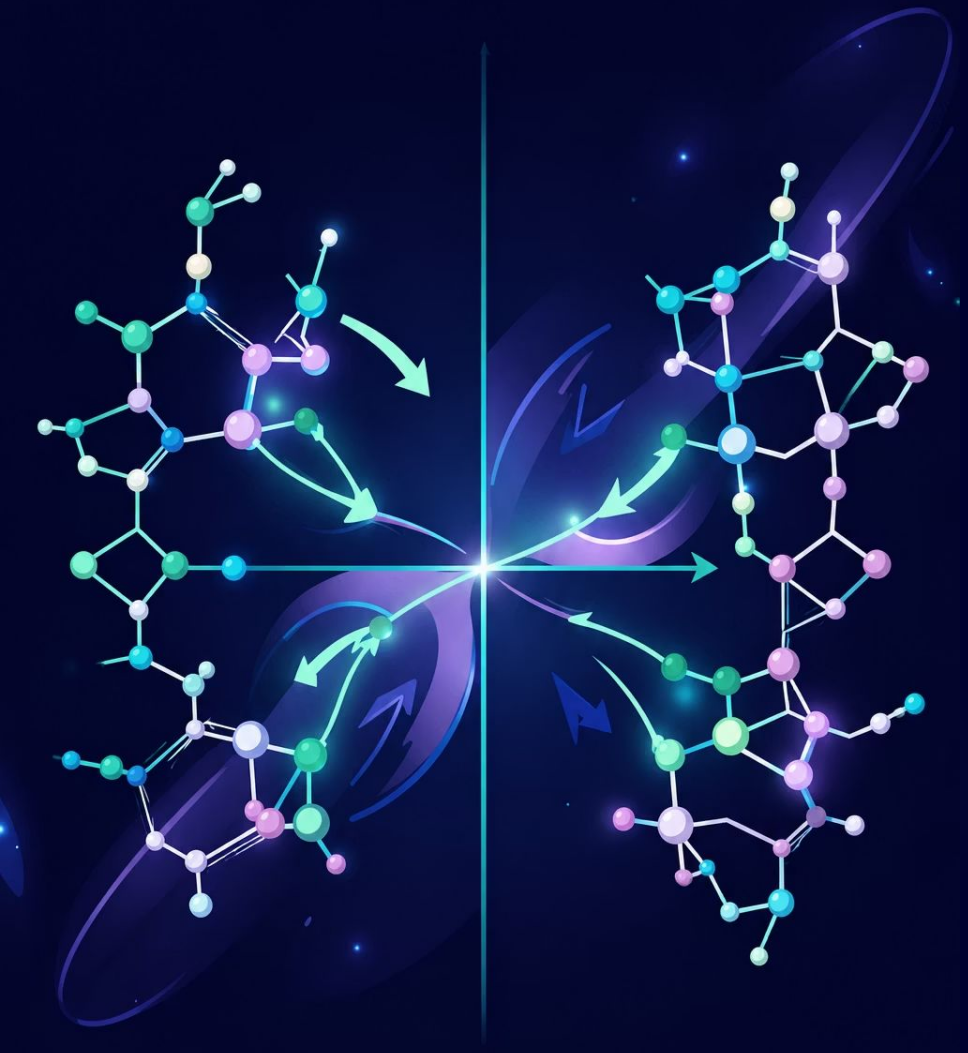


The Jafarpour Model

Stochastic Dynamics of a Modified Frank Model for Spontaneous Chiral Symmetry Breaking

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Motivation

Chiral Symmetry Breaking

How symmetry between left- and right-handed states is lost spontaneously.

Studying Homochirality

We want to inspect if homochirality can be obtained in absence of explicit chiral inhibition.

Experimental Evidences

Unlike the original Frank's Model which had no experimental evidence for numerous years barring one, the modified model was observed in recent ribozyme-mediated RNA replication experiments.

Reintroduction of the Decayed Specie

The first of the two Jafarpour models examined a model where we have no chiral inhibition however there is no way for to retrieve the once Decayed species either.

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Jafarpour Reaction Model

The new Jafarpour model introduced reversible reactions as opposed to a linear decay

Reaction Model:

$$A + D \xrightarrow{k_a} 2D$$

$$A + L \xrightarrow{k_a} 2L$$

$$A \underset{k_d}{\overset{k_n}{\rightleftharpoons}} D$$

$$A \underset{k_d}{\overset{k_n}{\rightleftharpoons}} L$$

⚠ Changes introuced

- A new parameter K_n was added to make sure we can reintroduce the species back into this medium.
- From the original franks model the chiral inhibition step was completely removed.

Mean field equations:

$$\frac{dA}{dt} = -k_a A(D + L) - 2k_n A + k_d(D + L)$$

$$\frac{dD}{dt} = k_a AD + k_n A - k_d D$$

$$\frac{dL}{dt} = k_a AL + k_n A - k_d L$$

The total concentration is conserved

$$A + D + L = N$$

This ensures we can have a fixed number of molecules in our system at any given time.

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Important Parameters

1. Chiral Order Parameter (ω)

$$\boxed{\omega = \frac{d - l}{d + l}}$$

The important limiting cases for this are:

$$\omega = 0 \qquad \text{for the racemic state}$$

$$\omega = 1 \qquad \text{for the } D\text{-homochiral state}$$

$$\omega = -1 \qquad \text{for the } L\text{-homochiral state}$$

2. Rate of change

We can define

$$\Delta = D - L$$

from solving the mean field equations we can obtain a relation

$$\frac{d\Delta}{dt} = k_a A(D - L) - k_d(D - L)$$

this will be especially useful for the analysis of various regimes later on

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Deterministic Analysis

Since there are 3 parameters, namely K_a, K_n, K_d we would want to examine their behavior and understand how this system behaves

Case 1:

We consider the modified reaction system K_a= K_n = K_d =1.0, N=1

$$A = 1 - D - L$$

We can substitute the value of A in the above equations and find the two rate equations in terms of D and L

$$\boxed{\frac{dD}{dt} = 1 - D - L - D^2 - DL}$$

$$\boxed{\frac{dL}{dt} = 1 - D - L - DL - L^2}$$

Equating both to 0

$$(L - D)(L + D) = 0$$

so we have two solutions, these are cases where we will find an equilibrium provided the nullclines of the system are present

$$\boxed{D = L}$$

$$\boxed{D = -L}$$

Since one of the two solutions is physically not feasible we will only take into account the $\boxed{D = L}$ solution for our calculations moving forward

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now we must find the two equilibrium points of the hence under the given condition we equate the nullcline putting the mean field equation to 0

we find two equilibrium points

$$\left(-\frac{1+\sqrt{3}}{2}, -\frac{1+\sqrt{3}}{2}\right) \quad \left(\frac{\sqrt{3}-1}{2}, \frac{\sqrt{3}-1}{2}\right)$$

since concentration can not be negative the negative roots are meaningless and we have only one useful solution. This point is the **RACEMIC EQUILIBRIUM** point of the system. Now we must perform a stability analysis via eigen decomposition to see how it behaves.

Construction of the Jacobian Matrix

$$J(D,L) = \begin{pmatrix} \frac{\partial f}{\partial D} & \frac{\partial f}{\partial L} \\ \frac{\partial g}{\partial D} & \frac{\partial g}{\partial L} \end{pmatrix}$$

we can expand this into

$$J(D,L) = \begin{pmatrix} -1-2D-L & -1-D \\ -1-D & -1-D-2L \end{pmatrix}$$

subsitutiting the values of D and L in the jacobian we obtain

$$J^* = \begin{pmatrix} \frac{1-3\sqrt{3}}{2} & -\frac{1+\sqrt{3}}{2} \\ -\frac{1+\sqrt{3}}{2} & \frac{1-3\sqrt{3}}{2} \end{pmatrix}$$

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Now that we have obtained the jacobian matrix at the desired equilibrium point we can now perform the stability test

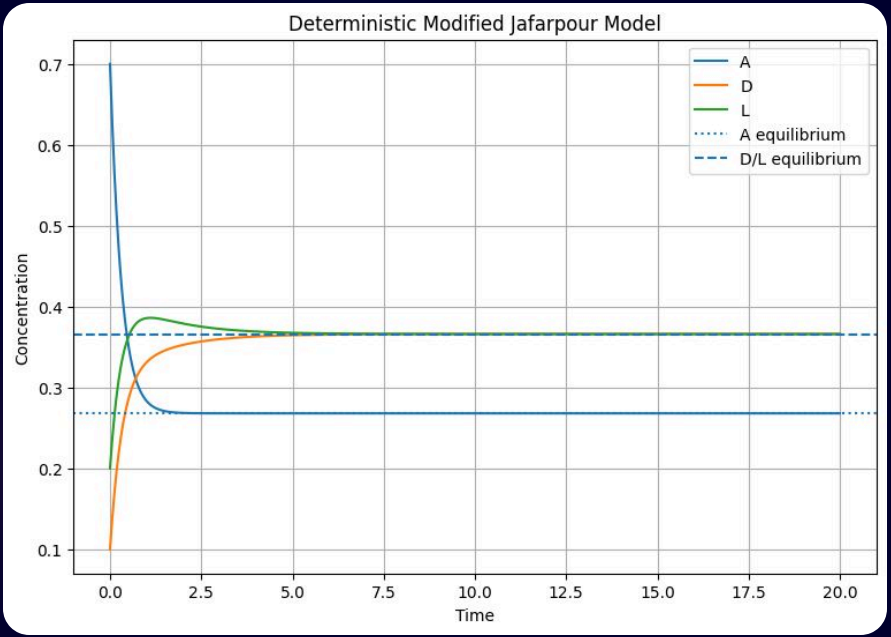
$$\begin{vmatrix} \frac{1-3\sqrt{3}}{2} - \lambda & -\frac{1+\sqrt{3}}{2} \\ -\frac{1+\sqrt{3}}{2} & \frac{1-3\sqrt{3}}{2} - \lambda \end{vmatrix} = 0$$

solving this determinant yields two eigen values

$$\boxed{\lambda_1 = -2\sqrt{3}}$$

$$\boxed{\lambda_2 = 1 - \sqrt{3}}$$

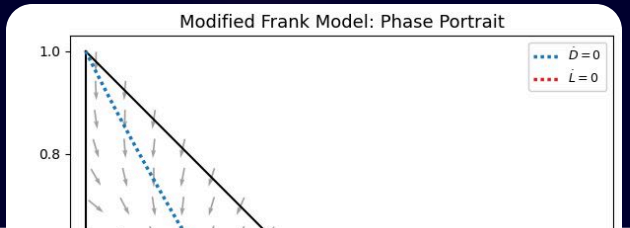
since all eigen values are negative we can safely say that this point would be **STABLE** and attract all trajectories.



Final concentrations:
A = 0.267949
D = 0.366025
L = 0.366025

Theoretical equilibrium:
A* = 0.267949
D* = 0.366025
L* = 0.366025

Phase Space Analysis



Conclusion:

A very important thing to observe here is unlike the previous models this has the ability to reintroduce a decayed specie back into the system, essentially if the concentration of a specie is higher it decays more, if its lesser it decays less. However they have an equal chance of being produced from the achiral precursor as it has no dep concentration. More over there is a very interesting phenomenon.

$$\frac{d(D-L)}{}$$

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If we happen to analyse the equation

$$\frac{d\Delta}{dt} = k_a A(D-L) - k_d(D-L)$$

we can make a few key observations, the amplification or suppression of the dominat specie is a proverbial tug of war between k_a and k_d hence we can make an obvious conclusion that homochirality can perhaps only be obtained when :

$$K_a A > K_d$$

This leads us to a very natural next step that is to look at regimes where we might see some amplification

Case 2:

We will now examine a case where $K_a = 1.0$, K_n =1.0 and K_d =0.0001 so essentially a lot of preference towards the autocatalysis

Now the modified mean nullcline equations are:

$$\boxed{\frac{dD}{dt} = 1 - 1.0001D - L - D^2 - DL = 0}$$

$$\boxed{\frac{dL}{dt} = 1 - D - 1.0001L - DL - L^2 = 0}$$

Equating both of them to 0 simultaneously again we obtain

$$\boxed{(L-D)(D+L+0.0001) = 0}$$

Hence we have 2 different solutions to this

$$\boxed{D = L} \quad \boxed{D = -0.0001 - L}$$

Since one of them is physically impossible $D = L$ is the only feasible solution to this system

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Stability analysis

We can use the condition to solve the nullclines, and after plugging the value we see that there is only one feasible solution at equilibrium

$$\boxed{(A^*, D^*, L^*) \approx (0.00003333,\ 0.4999833,\ 0.4999833)}$$

now after performing the stability test we for the 2D jacobian matrix wrt D and L we can observe that the eigen values are

$$\boxed{\lambda_1 \approx -3.00003333} \quad \boxed{\lambda_2 \approx -6.66674 \times 10^{-5}}$$

Since both eigen values are negative we have a stable fixed point here

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